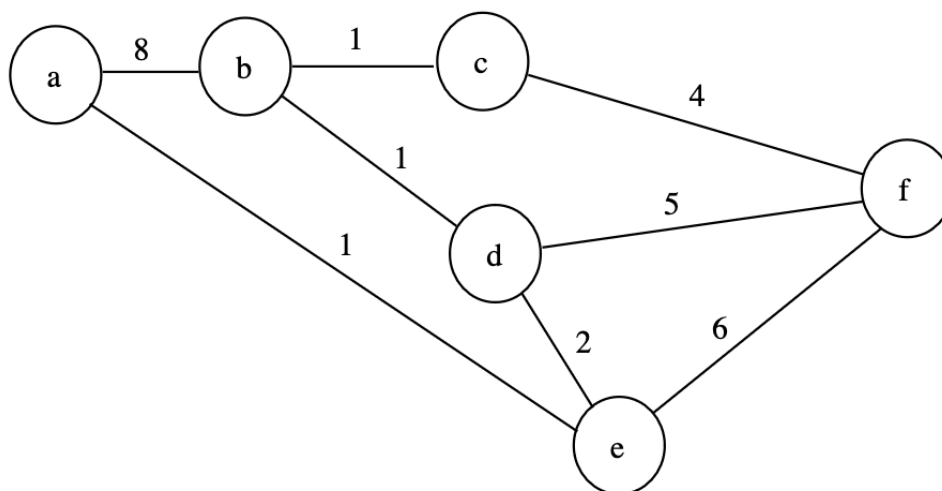


Question 1



a.

	$D^b(a)$	$D^c(a)$	$D^d(a)$	$D^e(a)$	$D^f(a)$
Initial	8	∞	∞	1	∞
1 exchange	8	∞	3	1	7
2 exchanges	4	∞	3	1	7
3 exchanges	4	5	3	1	7
4 exchanges	4	5	3	1	7
5 exchanges	4	5	3	1	7

b.

	$D^b(a)$	$D^c(a)$	$D^d(a)$	$D^e(a)$	$D^f(a)$
Initial	4	5	3	1	7
1 exchange	4	5	3	5	7
2 exchanges	4	5	5	5	8
3 exchanges	6	5	5	7	9
4 exchanges	6	7	7	7	9
5 exchanges	8	7	7	9	11

6 exchanges	8	9	9	11	11
7 exchanges	8	9	9	11	13
8 exchanges	8	9	9	11	13

c.

Next hop is shown next to the current cost so I can keep track of the next hops in each iteration.

	$D^b(a)$	$D^c(a)$	$D^d(a)$	$D^e(a)$	$D^f(a)$
Initial	4 d	5 b	3 e	1 a	7 e
1 exchange	4 d	5 b	3 e	∞	7 e
2 exchanges	4 d	5 b	12 f	∞	8 d
3 exchanges	8 a	5 b	∞	14 d/f	9 c
4 exchanges	8 a	9 b	9 b	15 c	9 c
5 exchanges	8 a	9 a	9 a	11 d	13 c
6 exchanges	8	9	9	11	13

d.

cost to send to group = (router c -> router e) + (router e -> router b) + (router e -> router f)
= 4 + 3 + 6 = 13

Question 2

M = link rate (bytes/s)

S = duration of maximum burst

R = tokens generation rate (token/s)

B = bucket capacity

M = 2 Gbps = 250,000,000 bytes/s

S = 1 ms = 0.001 s

R = 200 Mbps = 25,000,000 bytes/s = 25,000,000 tokens/s

$S = B / (M - R)$

$0.001s = B / (250\,000\,000 - 25\,000\,000)\text{bytes/s}$

$B = 0.001 \times 225\,000\,000 \text{ bytes/s}$

$B = 225\,000 \text{ tokens or } 1\,800\,000 \text{ bits}$

Question 3

a.

IP = 10.35.143.128

Binary = 00001010 00100011 10001111 10000000

Entry 3 (26 bit prefix)

00001010 00100011 10001111 10000000 & 11111111 11111111 11111111 11000000
= 00001010 00100011 10001111 10000000 = 10.35.143.128

10.35.143.128 != 10.35.143.0

Entry 4 (23 bit prefix)

00001010 00100011 10001111 10000000 & 11111111 11111111 11111110 00000000
= 00001010 00100011 10001110 00000000 = 10.35.142.0

10.35.142.0 == 10.35.142.0

An incoming datagram would be forwarded on interface A going to the address of the next hop router which is 10.35.128.1

b.

IP = 10.35.140.28

Binary = 00001010 00100011 10001100 00011100

Entry 3 (26 bit prefix)

00001010 00100011 10001100 00011100 & 11111111 11111111 11111111 11000000
= 00001010 00100011 10001100 00000000 = 10.35.140.0

10.35.140.0 != 10.35.143.0

Entry 4 (23 bit prefix)

00001010 00100011 10001100 00011100 & 11111111 11111111 11111110 00000000
= 00001010 00100011 10001100 00000000 = 10.35.140.0

10.35.140.0 != 10.35.142.0

Entry 2 (20 bit prefix)

00001010 00100011 10001100 00011100 & 11111111 11111111 11110000 00000000
= 00001010 00100011 10000000 00000000 = 10.35.128.0

10.35.128.0 == 10.35.128.0

An incoming datagram would be forwarded on interface A going to the address of the destination host's link layer address which is 10.35.140.28

c.

IP = 10.35.144.6

Binary = 00001010 00100011 10010000 00000110

Entry 3 (26 bit prefix)

00001010 00100011 10010000 00000110 & 11111111 11111111 11111111 11000000
= 00001010 00100011 10010000 00000000 = 10.35.144.0

10.35.144.0 != 10.35.143.0

Entry 4 (23 bit prefix)

00001010 00100011 10010000 00000110 & 11111111 11111111 11111110 00000000
= 00001010 00100011 10010000 00000000 = 10.35.144.0

10.35.144.0 != 10.35.142.0

Entry 2 (20 bit prefix)

00001010 00100011 10010000 00000110 & 11111111 11111111 11110000 00000000
= 00001010 00100011 10010000 00000000 = 10.35.144.0

10.35.144.0 != 10.35.128.0

Entry 1 (0 length prefix)

00001010 00100011 10010000 00000110 & 00000000 00000000 00000000 00000000
= 00000000 00000000 00000000 00000000 = 0.0.0.0

0.0.0.0 == 0.0.0.0

An incoming datagram would be forwarded on interface B going to the address of the next hop router which is 10.0.81.28

d.

IP = 10.35.143.6

Binary = 00001010 00100011 10001111 00000110

Entry 3 (26 bit prefix)

00001010 00100011 10001111 00000110 & 11111111 11111111 11111111 11000000
= 00001010 00100011 10001111 00000000 = 10.35.143.0

10.35.143.0 == 10.35.143.0

An incoming datagram would be forwarded on interface A going to the address of the next hop router which is 10.35.129.255